MSG No. 193.255 $P6_3/m'cm$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-6'2'm`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,3,5,10,11,12,13,14,15,20,22,24]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[2,4,6,7,8,9,16,17,18,19,21,23]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'.m`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,3,5,10,11,12,16,17,18,19,21,23]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[2,4,6,7,8,9,13,14,15,20,22,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,3,5,20,22,24]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[2,4,6,19,21,23]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[7,8,9,16,17,18]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[10,11,12,13,14,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,3,5,16,17,18]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[2,4,6,13,14,15]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[7,8,9,20,22,24]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[10,11,12,19,21,23]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,3,5,10,11,12]`
2	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[2,4,6,7,8,9]`
3	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[13,14,15,20,22,24]`
4	$[0,\, 0,\, -z]$	`[16,17,18,19,21,23]`

Table 6: Wyckoff site: `6f`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, 0\right]$	`[1,11,17,19]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,9,15,20]`
3	$\left[0,\, \frac{1}{2},\, 0\right]$	`[3,12,18,21]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[4,7,13,22]
5	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[5,10,16,23]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,8,14,24]

Table 7: Wyckoff site: 6g, site symmetry: m'2'm

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	[1,11,13,22]
2	$\left[x, x, \frac{3}{4}\right]$	[2,9,16,23]
3	$\left[0, x, \frac{1}{4}\right]$	[3,12,14,24]
4	$\left[-x, 0, \frac{3}{4}\right]$	[4,7,17,19]
5	$\left[-x, -x, \frac{1}{4}\right]$	[5,10,15,20]
6	$\left[0, -x, \frac{3}{4}\right]$	[6,8,18,21]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5]
2	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[2,4,6]
3	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[7,8,9]
4	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[10,11,12]
5	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[13,14,15]
6	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[16,17,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[19,21,23]
8	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[20,22,24]

Table 9: Wyckoff site: 12i, site symmetry: ..2'

No.	position	mapping
1	$[x, 2x, 0]$	[1,17]
2	$\left[-x, x, \frac{1}{2}\right]$	[2,15]
3	$[-2x, -x, 0]$	[3,18]
4	$\left[-x, -2x, \frac{1}{2}\right]$	[4,13]
5	$[x, -x, 0]$	[5,16]
6	$\left[2x, x, \frac{1}{2}\right]$	[6,14]
7	$\left[x, 2x, \frac{1}{2}\right]$	[7,22]
8	$\left[-2x, -x, \frac{1}{2}\right]$	[8,24]
9	$\left[x, -x, \frac{1}{2}\right]$	[9,20]
10	$[-x, x, 0]$	[10,23]
11	$[-x, -2x, 0]$	[11,19]
12	$[2x, x, 0]$	[12,21]

Table 10: Wyckoff site: `12j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	`[1,22]`
2	$\left[x-y,\, x,\, \frac{3}{4}\right]$	`[2,23]`
3	$\left[-y,\, x-y,\, \frac{1}{4}\right]$	`[3,24]`
4	$\left[-x,\, -y,\, \frac{3}{4}\right]$	`[4,19]`
5	$\left[-x+y,\, -x,\, \frac{1}{4}\right]$	`[5,20]`
6	$\left[y,\, -x+y,\, \frac{3}{4}\right]$	`[6,21]`
7	$\left[-x+y,\, y,\, \frac{3}{4}\right]$	`[7,17]`
8	$\left[-y,\, -x,\, \frac{3}{4}\right]$	`[8,18]`
9	$\left[x,\, x-y,\, \frac{3}{4}\right]$	`[9,16]`
10	$\left[-x,\, -x+y,\, \frac{1}{4}\right]$	`[10,15]`
11	$\left[x-y,\, -y,\, \frac{1}{4}\right]$	`[11,13]`
12	$\left[y,\, x,\, \frac{1}{4}\right]$	`[12,14]`

Table 11: Wyckoff site: `12k`, site symmetry: `..m`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,11]`
2	$\left[x,\, x,\, z+\frac{1}{2}\right]$	`[2,9]`
3	$[0,\, x,\, z]$	`[3,12]`
4	$\left[-x,\, 0,\, z+\frac{1}{2}\right]$	`[4,7]`
5	$[-x,\, -x,\, z]$	`[5,10]`
6	$\left[0,\, -x,\, z+\frac{1}{2}\right]$	`[6,8]`
7	$\left[x,\, 0,\, \frac{1}{2}-z\right]$	`[13,22]`
8	$\left[0,\, x,\, \frac{1}{2}-z\right]$	`[14,24]`
9	$\left[-x,\, -x,\, \frac{1}{2}-z\right]$	`[15,20]`
10	$[x,\, x,\, -z]$	`[16,23]`
11	$[-x,\, 0,\, -z]$	`[17,19]`
12	$[0,\, -x,\, -z]$	`[18,21]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$[-y,\, x-y,\, z]$	`[3]`
4	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[-x+y,\, -x,\, z]$	`[5]`
6	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	`[7]`
8	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[8]`
9	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	`[9]`
10	$[-x,\, -x+y,\, z]$	`[10]`
11	$[x-y,\, -y,\, z]$	`[11]`
12	$[y,\, x,\, z]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[14]
15	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	[15]
16	$[x,\ x-y,\ -z]$	[16]
17	$[-x+y,\ y,\ -z]$	[17]
18	$[-y,\ -x,\ -z]$	[18]
19	$[-x,\ -y,\ -z]$	[19]
20	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[20]
21	$[y,\ -x+y,\ -z]$	[21]
22	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[22]
23	$[x-y,\ x,\ -z]$	[23]
24	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[24]